Question Paper

Exam Date & Time: 27-Nov-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

SEVENTH SEMESTER B.TECH END SEMESTER EXAMINATIONS, NOV/DEC 2024

INTERNET OF THINGS [ECE 4054]

Marks: 50 **Duration: 180 mins.**

Answer all the questions. Section Duration: 180 mins

Instructions to Candidates: Answer ALL questions Missing data may be suitably assumed

1)		Explain the IoT system architecture in terms of four stages and explain the functions of each layer in detail.	(4)
	A)		
	B)	Draw the IoT protocol suite in comparison with ISO: OSI, TCP/IP reference models. Describe the features of each layer of the IoT protocol suite.	(4)
	C)	Differentiate the features of MANETs, VANETs, and UAV networks.	(2)
2)		Describe the working principle of NFC with relevant diagrams and discuss the modes of operations.	(4)
	A)		
	B)	Explain the features of IEEE 802.15.4 with relevant diagrams.	(4)
	C)	Discuss the types of interoperability in IoT.	(2)
3)		With relevant diagrams, describe the working principle of OpenFlow protocol.	(4)
	A)		
	B)	Discuss the features of various service models in cloud computing.	(4)
	C)	What is Big data? Explain its characteristics in detail.	(2)
4)		Explain the working of sensor cloud architecture with relevant diagrams. Mention the differences between sensor cloud and wireless sensor networks with respect to various attributes.	(4)
	A)		
	B)	Why is fog computing essential? With help of relevant diagrams, discuss the working of fog for various levels of time-sensitive data.	(4)
	C)	What are the features of various types of clouds?	(2)
5)		Describe any five quantitative analysis techniques in data analytics.	(4)
	A)		
	B)	Explain the features of smart grid over traditional electrical grid. Discuss the benefits of smart grid for consumers and stakeholders.	(4)

C) Brief any FOUR key application areas of IIoT. (2)

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